more tracers can be enabled during compilation, as per requirements. The listed tracers are for debugging. Once the kernel compilation is complete, and you have booted to the new kernel, tracing can be initiated.

Tracing

Files in the tracing directory (*/sys/kernel/debug/tracing*) control the tracing ability (refer to Figure 2 for a list of files). A few files could be different, depending upon what tracers you selected during kernel configuration. You can obtain information on these files from the *<kernel source>/ Documentation/tracing* directory. Let's explore a few of the important ones:

- *available_tracers*: This shows what tracers are compiled to trace the system.
- *current_tracer*: Displays what tracer is currently enabled. Can be changed by echoing a new tracer into it.
- *tracing_enabled*: Lets you enable or disable the current tracing.
- *trace*: Actual trace output.
- *set_ftrace_pid*: Sets the PID of the process for which trace needs to be performed.

To find out the available tracers, just *cat* the *available_ tracers* file. Tracers in the space-separated output include: *nop* (not a tracer, this is set by default); *function* (function tracer); *function_graph* (function graph tracer), etc:

```
# cat available_tracers
blk function_graph mmiotrace wakeup_rt wakeup irqsoff function
sched_switch nop
```

Once you identify the tracer that you want to use, enable it (Ftrace takes only one tracer at a time):

```
# cat current_tracer # to see what tracer is currently in use.
# echo function > current_tracer # select a particular tracer.
# cat current_tracer # check whether we got what we wanted.
```

To start tracing, use the following commands:

```
# echo 1 > tracing_enabled  # initiate tracing
# cat trace > /tmp/trace.txt # save the contents of the trace to
a temporary file.
# echo 0 > tracing_enabled  # disable tracing
# cat /tmp/trace.txt         # to see the output of the trace
file.
```

The trace output is now in the trace.txt file. A sample output of a function trace obtained with the above commands is shown in Figure 3.

To be continued

We will explore more Ftrace options, and consider some tracing scenarios in the second part of this series, next month. END

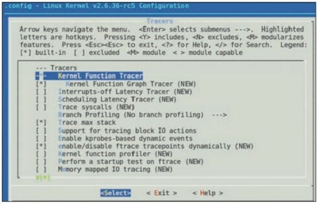

Figure 1: Kernel configuration options for tracing

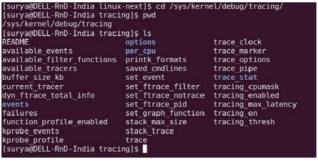

Figure 2: Tracing files

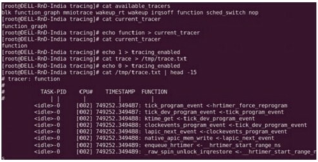

Figure 3: Sample trace output

References

Linux kernel's documentation/tracing directory has been referred to. Apart from that, a few articles from Lwn were referred to, as well. Readers can find abundant information in these resources for additional information.

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